Formative Assessment and Technology: Reflections Developed through the Collaboration between Teachers and Researchers

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Abstract In this work, we present the analysis of the ways in which formative assessment processes can be developed, by the teacher and the students, thanks to the support given by technology. The analysis is carried out focusing on two case studies developed in France and Italy within the European Project FaSMEd, with two main aims: (1) highlighting how the different functionalities of technology could support formative assessment strategies at the teacher's, the students' and the peers' levels; and (2) characterising the dynamics that intervene within programs involving a strict collaboration between teachers and researchers. Through the analysis of the two case studies we discuss, on one side, the effectiveness of the adopted theoretical tools, and, on the other side, the contribution, in terms of professional development, of the collaborative work developed within the project.

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Introduction

In this chapter we analyse the role played by technology in supporting the formative assessment (FA) process, referring to some examples from the case studies developed in France and Italy within the European Project FaSMEd (Improving progress for lower achievers through Formative Assessment in Science and Mathematics Education).

The FaSMEd project aims at investigating the role of technologically enhanced FA methods in raising the attainment levels of low-achieving students. It relies on the hypothesis that a technological environment can potentially support both the students and the teachers in getting information about students' achievement in real-time. Indeed, since connectivity and feedback are enhanced, students can use data from technology for informing their learning trajectories, and teachers have the possibility of collecting data from students, making more timely formative interpretations and informing their future teaching.

In line with this hypothesis, the project investigates: (a) students’ use of FA data to inform their learning trajectories; (b) teachers’ ways of processing FA data from students using a range of technologies; (c) teachers’ ways of using these data to inform their future teaching; and (d) the role played by technology, as a learning tool, in enabling the teachers to become more informed about student understanding.

The research is based on successive cycles of design, observation, analysis and redesign of classroom sequences (Swan 2014) in order to produce and feed into the *toolkit*, a set of curriculum materials and methods for pedagogical intervention aimed at supporting the development of practice.

The teachers engaged in the project are involved in the different phases of design, implementation, analysis and subsequent redesign and adaptation of the toolkit. Besides FA, technology and low-achievers, the FaSMEd project addresses also the issue of teacher professional development, as a consequence of the collaboration between teachers and researchers.

Some studies highlighted the considerable amount of time for teachers to change their beliefs about teaching and learning, classroom culture and the teacher’s role (Foshayla and Bellman 2012) and their ways of being so that FA with technology becomes an integral part of their practice (Feldman and Capobianco 2008).

Our hypothesis (FaSMEd DOW, p. 8 PART B[2]) is that, to enable teachers to incorporate elements of the new pedagogical model into their everyday teaching, it is important to give them the opportunity to engage in a process of development

[2] FaSMEd: Improving progress for lower achievers through Formative Assessment in Science and Mathematics Education. Annex I, Description of Work.

where they can reflect on and contrast their experience in using this approach. The teachers are therefore engaged as practitioner researchers since they undertake inquiry-based practice, strictly collaborate with researchers, and engage seriously with reflective developmental practice.

Considering the double concern of the project this chapter aims at (1) highlighting how the different functionalities of technology could support formative assessment strategies at the teacher's, the students' and the peers' levels; and (2) characterising the dynamics that intervene within programs involving a strict collaboration between teachers and researchers. We present the theoretical framework, which is constituted by two main poles: one frames the role of technology in supporting FA, and the other provides tools to analyse the dynamics within a teachers-researchers collaboration. Then we will present two case studies, from teaching experiments carried out in France (Lyon) and in Italy (Torino). The presentation of the case studies will include methodological aspects as well as data analysis from the classroom. We will conclude proposing some reflections arisen from the comparison of our case studies.

Theoretical framework

FA and the role of technology

In the FaSMEd Project, FA is intended as a method of teaching where

> [...] evidence about student achievement is elicited, interpreted, and used by teachers, learners, or their peers, to make decisions about the next steps in instruction that are likely to be better, or better founded, than the decisions they would have taken in the absence of the evidence that was elicited (Black and Wiliam 2009, p. 7).

Such learning evidences can be collected, interpreted and exploited in different moments of the learning process and with different purposes. In particular, Wiliam and Thompson (2007, adapted from Ramaprasad 1983) focus on three central processes in learning and teaching: (a) Establishing where learners are in their learning; (b) Establishing where learners are going; (c) Establishing how to get there.

Different agents are involved in these three processes: the teacher, the learners and their peers. Black and Wiliam (2009) conceptualise FA as consisting of five key strategies, that could be activated by agents:

1. Clarifying and sharing learning intentions and criteria for success;
2. Engineering effective classroom discussions and other learning tasks that elicit evidence of student understanding;
3. Providing feedback that moves learners forward;
4. Activating students as instructional resources for one another;
5. Activating students as the owners of their own learning.

Effective feedback from the different involved agents plays a central role in FA. According to Hattie and Temperley (2007), there are four major levels, and the level at which feedback is produced influences its effectiveness. They distinguish between:

1. feedback about the task, which includes feedback about how well a task is being accomplished or performed;
2. feedback about the processing of the task, which concerns the processes underlying tasks or relating and extending tasks;
3. feedback about self-regulation, which addresses the way students monitor, direct, and regulate actions toward the learning goal;
4. feedback about the self as a person, which expresses positive (and sometimes negative) evaluations and affect about the student.

Recently, research has started to investigate the different ways in which technology could support FA. Quellmalz and colleagues (2012), for example, stresses that technology enables the assessment of those aspects of cognition and performance that are complex and dynamic, through rich and authentic contexts, interactive and dynamic responses, individualized feedback and coaching, and diagnostic progress reporting.

Referring specifically to *connected classroom technologies*[3], further aspects are pointed out as features that make them effective tools for FA:

1. they may enable the teachers to monitor students' incremental progress and keep them oriented on the path to deep conceptual understanding, providing appropriate remediation to address student needs (Irving 2006, Shirley et al. 2011);
2. they may support positive students' thinking habits, such as arguing for their point of view (Roschelle et al. 2007), creating immersive learning environments that highlight problem-solving processes (Looney 2010) and giving powerful clues to what students are doing, thinking, and understanding (Roschelle et al. 2004);
3. they may enable most or all of the students contribute to the activities and work toward the classroom performance, taking a more active role in the discussions (Roschelle and Pea 2002; Shirley et al. 2011);
4. they may provide students with immediate private feedback, encouraging them to reflect and monitor their own progress (Looney 2010; Roschelle et al. 2007);
5. they may enable to carry out a multi-level analyses of patterns of interactions and outcomes thanks to their potential to instrument the learning space to collect the content of students' interaction over longer timespans and over multiple sets of classroom participants (Roschelle and Pea 2002).

[3] "Connected classroom technology" refers to a networked system of personal computers or handheld devices specifically designed to be used in a classroom for interactive teaching and learning (Irving 2006).

Together with the other researchers involved in the FaSMEd project, we started developing a three-dimensional framework[4] that extends Black and Wiliam's (2009) model to include the use of technology in FA processes. The new model takes into account three main dimensions:

- the five FA key-strategies,
- the three main agents (teacher, student, peers/group),
- the functionalities through which technology can support the three agents in developing the FA strategies.Within the third dimension (functionalities of technology), that we have introduced, we distinguish three main categories, according to the different uses of technology for FA within the FaSMEd project:
  - *Sending and displaying*: when technology is used to support communication among the agents of FA processes and to activate fruitful discussions. For example: sending questions and answers, sending messages, sending files, displaying and sharing screens to the whole class or to specific students, sharing students' worksheets.
  - *Processing and analysing*: includes all the functionalities that support the processing and the analysis of the data collected during the lessons, such as the statistics of students' answers to polls or questionnaires, the feedbacks given directly by the technology to the students, the tracking of students' learning paths.
  - *Providing an interactive environment*: those functionalities of technology that enable to create a shared interactive environment within which students can work individually or collaboratively on a task or a learning environment where mathematical/scientific contents could be explored.

The following chart outlines the FaSMEd three-dimensional framework, where the three dimensions described above are represented along the axes of a three-dimensional Cartesian reference (see Fig. 1).

In the analysis of our examples, we will use this three-dimensional framework to describe and characterize the use of the technology to support FA in classrooms. In the discussion of each example we will also use other theoretical references introduced in this section (i.e. the levels of feedback) to go into detail in the analysis.

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[4] The framework was initially developed during a meeting in Essen (Germany) in July 2015 by some of the FaSMEd partners from Italy, United Kingdom, and Germany. It was also presented at ECER 2015, in Budapest, during the symposium "Formative Assessment in Science and Mathematics Education".

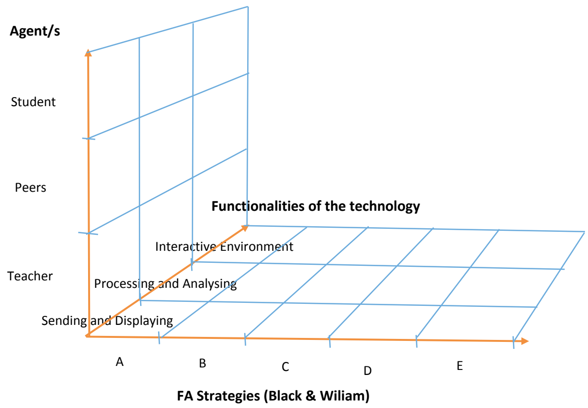

Fig. 1 Chart of the FaSMEd three-dimensional framework:

A. Clarifying and sharing learning intentions and criteria for success;
B. Engineering effective classroom discussions and other learning tasks that elicit evidence of student understanding;
C. Providing feedback that moves learners forward;
D. Activating students as instructional resources for one another;
E. Activating students as the owners of their own learning.

Meta-Didactical Transposition

In tune with the model of design-based research (Shavelson et al. 2003) we take into account the existing state of teachers' pedagogical knowledge relatively to formative assessment and use of technology and the evolution of this knowledge as a result of the collaboration between teachers and researchers. In order to analyse the complex dynamics involved in these processes, we refer to the model of the Meta-Didactical Transposition (MDT) developed by Arzarello and colleagues (2014).

The MDT model is based on the Anthropological Theory of the Didactic (ATD, Chevallard 1985, 1999) in which every human activity within an institution can be described by a *praxeology*. A praxeology consists of the *praxis,* the *"how to",* the techniques allowing to solve tasks that can be classified into types of tasks, and the *logos,* that is to say the discourses on these techniques bringing their justification in reference to a certain theory. A praxeology is therefore the given quadruplet type of task, techniques, justification of the techniques and theory developed within specific institutions. In line with this perspective, the MDT-model considers teacher education as a human activity that takes place in an (or several) institution(s).

During the work with teachers, the research team develops tasks and techniques in order to solve these tasks at a didactical level but also builds justifications of these techniques at a teaching level of reflection. These praxeologies are built at a meta-level justifying the term of meta-didactical praxeologies. But they are also built and discussed for applications in the classroom that is to say at the didactical level. In this way, two dialectics are generated. A *first dialectic* is developed at a didactical level in the classroom between students, teacher and knowledge. A *second dialectic* is developed at a meta-didactical level in the interaction between teachers and researchers relatively to the interpretation of the first dialectic.

> Typically the second (meta-didactical) dialectic arises from a contrast/comparison between the researchers' praxeologies and the teachers' praxeologies and the first dialectic engenders the second one as an outcome of a suitable meta-didactical trajectory, which is designed by the researchers. It is through this double dialectic that teachers' and researchers develop a shared praxeology. (Aldon et al. 2013, p.104)

In the FaSMEd project, the general discussion about FA at a meta-level, for example using the three-dimensional model, nurtures the construction of activities for the classroom and their implementation in the classrooms give feedback that makes the model evolve and participate to the teachers' professional development. During the FaSMEd experimental phases (task design, a-posteriori reflections, etc.), ideally researchers share research results and innovative inputs, while teachers mainly offer their professional knowledge and the pragmatic justifications of these practices. In this way, during the MDT process, the researchers' praxeologies encounter the teachers' ones, and it may happen that components of praxeologies that were *external* to a certain community become gradually *internal*, within a praxeology *shared* by the two communities. This process, called internalisation, can be iterated in a cyclic way.

Case studies

FaSMEd in France: one example and its analysis

Amongst all studies done in the context of FaSMEd in France, we choose to highlight one particular sequence in a grade 9 class where all students are equipped with tablets that are connected within a network.

Our methodological choice consists in giving to teachers the responsibility for designing lessons. We ask them to fill in a grid concerning important points to reflect upon before and after an observed lesson. We collect this information in order to contextualise what is happening in the classroom when we are about to visit it. While observing lessons, we collect videos, pictures, audios, teacher's report and notes. We also have intermediary discussions and meetings with the teacher, and final interviews with the teacher and with the students. The object of our analysis are the dynamics within the FaSMEd three-dimensional model (see above), which represent the agents' different uses of the technology for enhancing the process of formative assessment.

In order to account for the mathematical knowledge and competences at stake, we refer to the *Theory of Didactic Situation* (TDS, Brousseau 1998). Starting from Brousseau's didactic triangle to interpret the mutual relationships between teacher, student and knowledge, we consider its tridimensionalisation. Indeed, with the introduction of the technological dimension in the classroom, this triangle becomes a didactic tetrahedron, where technology represents a new vertex establishing relationships with the other didactic actors of the mathematical situation and getting a part of the *milieu* that teachers and students have to cope with.In the tetrahedron, the edges connect the vertices in order to describe and to analyse the mutual relationships between the four actors of the didactic situation: student, teacher, knowledge, technology. A face of the tetrahedron represents instead a point of view on the situation taking into account the relationships of three of the four actors. On the contrary, in the FaSMEd three-dimensional model, the axes correspond to agents, FA strategies and functionalities of technology. Thus, the relationships agents-technology or agents-knowledge can be found within the three-dimensional model, when for example an agent uses technology or mobilises knowledge in a particular moment of the FA process. For example, in Fig. 2, we can analyse the mathematical situation in which the teacher is using technology for processing and analysing data from the classroom, in order to provide feedback to students. We are interested in the evolution of the didactic situation described by the tetrahedrons within the cuboids.

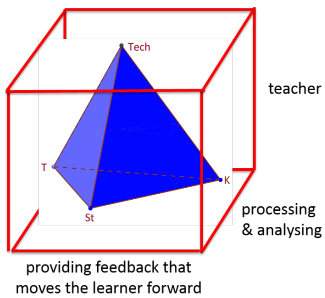

Fig. 2 The didactic situation within a particular cuboid of the FaSMEd three-dimensional model

In this grade 9th class and for networking tablets, the teacher (Thomas) uses the NetSupport School software that allows classroom monitoring, management, orchestration and collaboration. In addition, Thomas decides to use MapleTA that is an online testing and assessment system designed especially for courses involving mathematics. The classroom is also equipped with an IWB. From a didactical point of view, the use of such technologies for formative assessment in his classroom is completely new for the teacher. These aspects of professional development will be further analysed below under the lens of the Meta-Didactical Transposition.

The case study leans on a sequence about linear functions, where the following competences are to be acquired, according to the different representations of functions.

a. Calculating and detecting images.
b. Calculating and detecting inverse images.
c. Recognising a linear function.
d. Shifting from the graphical frame to the algebraic frame and vice versa.

Thomas decides to create a sequence of questionnaires around these four competences, using MapleTA. Following a typical Thomas' teaching sequence with MapleTA, we propose to analyse three specific episodes taken from our observations and referred to the third quiz proposed by Thomas to the students during this learning sequence about linear functions. The first moment concerns a student taking the quiz and the teacher declaring his potential use of the class' results. In the second episode, the teacher comments the quiz results of a student and, during the third excerpt, the teacher comments the whole set of the class' results.

First episode

Mor is working on Question 3 concerning the competence (a): calculating and detecting images (see Fig. 3). Formulated in the graphical register of representation, the question is: "The curve below represents a linear function. The image of 9 is -2. True/False."

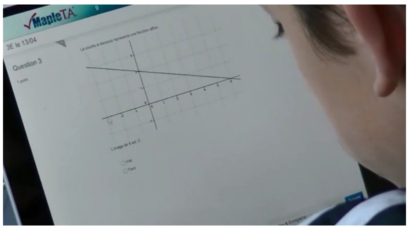

Fig. 3 Mor is reading Question 3 on his tablet on the platform MapleTA

Mor is a low-achiever, with high difficulties in mathematics. Since he is working alone on the mathematical task, he is *activated as the owner of his own learning* (E). He is reading the task given by the teacher on MapleTA. Therefore, the technology is used with the functionality of 'sending and displaying'. The student faces the didactic situation devolved by the teacher, represented by the tetrahedron in Fig. 4.

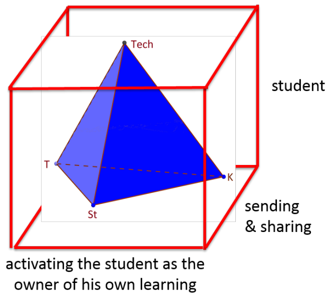

Fig. 4 The student is activated as owner of his own learning in the process of sending and displaying started with MapleTA by the teacher

After a while, Mor copies the question, leaves MapleTA, and pastes the question on the interactive environment of his tablet (OneNote) in order to work on the given graphical representation using his previous experience on similar exercises. On his screen, indeed, we can see a previously solved exercise that is very similar to the new one (Fig. 5a). Mor starts using the same graphical technique (Fig. 5b), mobilising his knowledge as a reflexive student. He has transformed the didactic situation into an a-didactic situation represented by the face Student-Knowledge-Technology of the tetrahedron, where he acts on a reacting *milieu*.

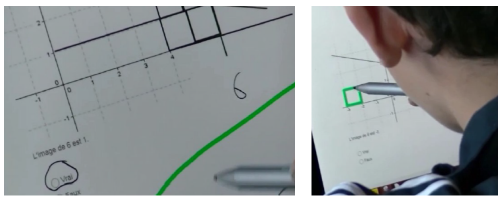

Fig. 5a and 5b Mor working on OneNote

Confronted to the *milieu*, composed of the task, the technology and the previous knowledge, Mor reflects on the mathematical situation mobilising his knowledge and finally submits his answer, sending it back to the teacher through the MapleTA. At the student's level, there is a shift between different functionalities of technology (Fig. 6), and the devolution of the mathematical situation by the teacher is at the base of this dynamics.

The back and forth between the two cuboids in Fig. 6, related to the different functionalities of technology, is the starting point of a more extended dynamics:

- towards other FA strategies;
- towards the peers and the teacher;
- towards other functionalities of technology.

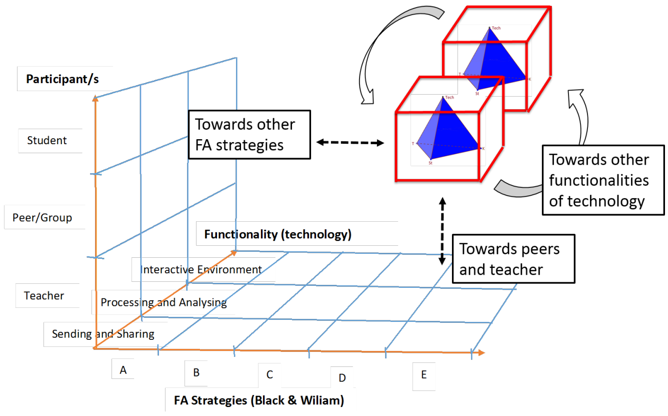

Fig. 6 Dynamics within the three-dimensional model at the student's level between the functionalities "Sending and Displaying" and "Providing an interactive environment" are the starting point of other dynamics

Moving on to the teacher's level, he is collecting the students' answers and will use MapleTA for 'processing and analysing' such data. To this purpose, during the lesson, the teacher declares his potential FA strategies depending on the students' responses.

> Thomas (to Ant, an other student): I don't know if I'm going to take it into account or not. The idea is that I would like to mark it. If I realise that it doesn't work… I don't know… I'm going to see what's going on… At least I'll know that you don't succeed here. You can skip it if you don't know what to do.

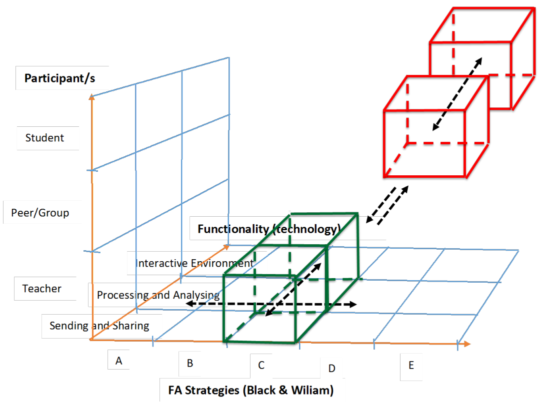

Fig. 7 Dynamics between the individual students and the teacher are the driving force of the dynamics at the teacher's level between the functionalities "Sending and Displaying" and "Processing and Analysing" for providing feedback to students

The students' results are feedback for the teacher, who will process and analyse these data. Depending on the students' performances, he may adapt his teaching, for example by choosing the FA strategy of providing feedback to students (Fig. 7). Through his words, Thomas gives to Ant a '*feedback about the task'*, by saying "At least I'll know that you don't succeed here".

Second episode

Teacher's feedback can be made on the spot, like in the second transcription that we propose to analyse. Rom has completed his quiz, submitted his answers and got a '*feedback about the task*' from MapleTA: "good answer" or "wrong answer". Then he calls Thomas in order to have further explanations.

> Thomas: The first one is right, the second one is false, the third one is right, and the fourth one is false. Finally, I consider that you were right on the two that are easier to explain and you got false on the two that require more mathematical work. That's normal. I consider your result as normal.

Both the teacher and the student benefit from the feedback in this episode. The student gets a '*feedback about the processing of the task'* and also on his global performance according to the teacher's norm. The teacher, who analyses this quiz result on the spot and considers it as normal, gets information about the student's achievement.

Third episode

Teacher's feedback can be made after reflection and this is the case of the third transcription. When all students have completed the quiz, Thomas leads the correction of the questions with the whole classroom using NetSupport School. The lesson after, he proposes a global lecture of the class' results at the three quiz and analyses the whole set of answers stored by MapleTA, by showing them at the IWB and commenting them with the students. In this way, he provides feedback for the whole class on the attained mathematical competences.

> Thomas: [Here are your results] on several trials. What we can see is that in calculating images you reached 0.778. What does it mean? […] about 8 successful students over 10, here. There we had 6 over 10, then 8 over 10. So we are good in calculating images. […] I'm not going further. However, we'll come back on determining the expression of a linear function. 0.1, you see 0.1, 0.3, and here we went down at 0.2. […] I would like to get to realise if I succeed in teaching you two or three things last time, so we are going to work again on these two questions. Open Maple TA, and answer the two questions of the day. Let's go.

Thomas analyses the class' results and he *clarifies the learning intentions and criteria for success*. He has worked again with the students on the required competences during the correction phase, and now he wants to test again the competences revealed as not achieved by the analysis of the results, namely competences (b) and (d). Thus, he *engineers other learning tasks* on MapleTA. Two new questions are properly prepared and sent to students as a result of this dynamics (Fig. 8). From Thomas' words, we can observe that, as he expected in the first episode, he has adapted his teaching depending on students' progressive achievement.

More generally, relatively to FA strategies, Thomas orchestrates the different functionalities of technology in direction of individual students, of the whole classroom or even of himself. Indeed, moving from 'processing and analysing' data to 'sending and displaying' results or new learning tasks allows him to choose the most powerful FA strategy according to students' mastering of the competences at stake.

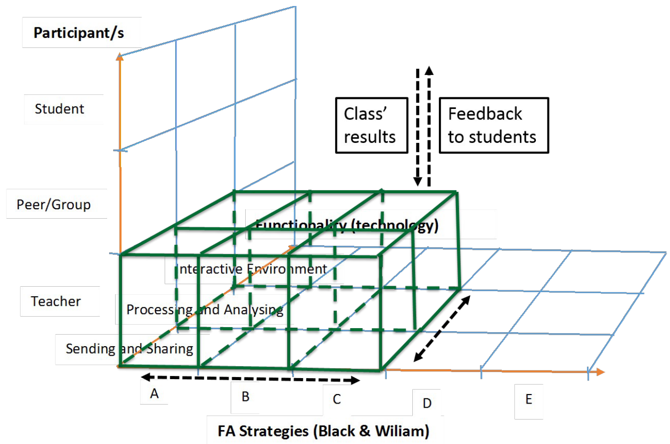

Fig. 8 Dynamics at the teacher's level to provide feedback to the whole class

FaSMEd in Italy: one example and its analysis

In Italy the FaSMEd Project involves 19 teachers from three different clusters of schools located in the North-West of Italy. 12 of them work in primary school (grades 4-5) and the other 7 in lower secondary school (grades 6-7). All the teachers work on the same mathematical topic: functions and their different representations (symbolic representation, tables, graphs).

We believe that low achievement is linked not only to a lack of basic competences, but also to metacognitive factors. For this reason, during class activities, it is important to make students: (a) develop ongoing reflections on the teaching-learning processes; (b) make their thinking visible (Collins et al. 1989) and share it with the teacher and the classmates.

Starting from these assumptions, we chose to exploit IDM-TClass, a *connected classroom technology,* because it enables both to share the students' ongoing and final productions, and to collect their opinions during and after the activities. Specifically, it allows the teacher to: (a) show, to one or more students, the teacher's screen and also other students' screens; (b) distribute documents to students and collect documents from the students' tablets; (c) create different kinds of tests and have a real-time visualization of the correct and the wrong answers; (d) create instant polls and immediately show their results to the whole class.

Each school has been provided with tablets for the students (who work in pairs), computers for the teachers and, where the interactive whiteboard was not available, a data projector to display students' written productions. The students' tablets are connected with the teachers' laptop through the IDM-TClass software. IDM-TClass was integrated within a set of activities coming from different sources. Among them, the ArAl Units, which are models of sequences of didactic paths developed within the project "ArAl – Arithmetic pathways towards favouring pre-algebraic thinking" (Cusi et al. 2011).

For each lesson carried out in the classes, we have prepared a set of different worksheets, aimed at:

- supporting the students in the verbalisation and the representation of the relations introduced within the lesson;
- enabling the students to compare and discuss their answers;
- making the students reflect at both the cognitive and metacognitive level.

In this paragraph we are going to analyse an excerpt from a class discussion, which was chosen because of the richness in the levels of feedbacks provided and in the FA strategies that are activated. The discussion refers to the following worksheet:

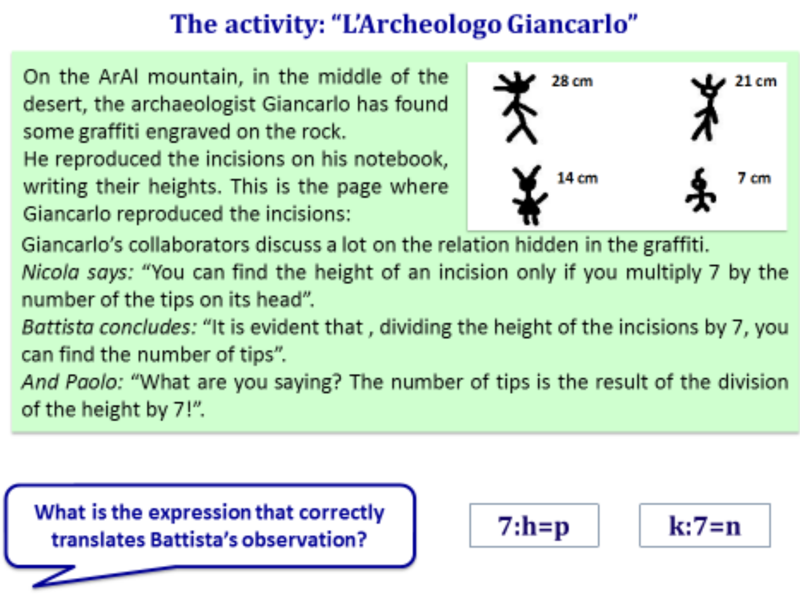

Fig. 9 The worksheet provided to students

We remark that this is not the first worksheet on this problem. During the previous lesson, students were asked to discuss and compare Nicola, Battista and Paolo's statements and to interpret a symbolic expression (7 x n = k) proposed by a fictive student from another class with reference to the problem.

During the lesson reported in this example, 5th grade students are asked to answer the question through a poll. The exploited functionality of the technology is 'processing and analysing', because the technological tool collects all the students' answers and processes them, displaying an analytical as well as a synthetic overview (bar chart) to the teacher. We (the teacher and the researchers) decided not to provide an immediate correction.

When all the students answer the question, the teacher (Monica) shares with them her screen, where the bar chart and the list of students' answers are displayed:

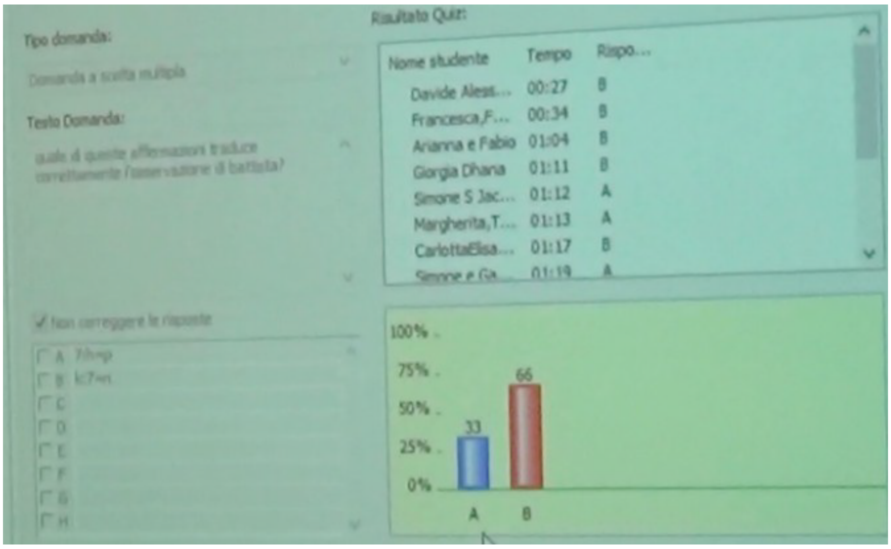

Fig. 10 The teacher's screen shared with the students

The worksheet is also projected on the interactive whiteboard, next to the poll.

Monica observes that the 33% of the students chose the answer *7:h = p*, while the 66% of the students chose the answer *k:7 = n*. Monica reads the list of the couples of students and the corresponding answer.

Monica chooses not to tell to the students what the right answer is, and asks to the different pairs to explain why they chose a specific answer. The class discusses on the possible strategies that could be used to identify the correct expression, in case the only reading of Battista's observation is not enough. The students are invited to check if the number of tips and the height of every incision verify the two expressions. Some students are asked to substitute, in the two expressions, the different values connected to each incision (4,28; 3,21; 2,14; 1,7). One of them observes that she discarded expression A because the result of the division 7:28 is not 4.

Alice, softly, says that *7:7 = 1*. Monica asks her to explain what she means. We report the related excerpt:

1. Monica (*to Alice*): "What were you saying?" 2. Alice: "I was saying that, for example, the figure, the one on the bottom right, is 7cm, so *7:7* is *1*, therefore the result is not a decimal number, while with the others (*the other figures*) it is (*the result is a decimal number*).
Monica focuses on Alice's observation and states that the chosen expression should represent all the incisions, not only the first one. Lisa and Nicolò ask if they can change their mind.
9. Monica: "Have you changed your mind? That is, Lisa, you chose answer A, but now you have changed your mind. Why?" 10. Lisa: "Ahem … *7* is only that figure. While, if you divide the height by 7, you mean all the figures."
Another student declares that, although h in Italy always stands for the height, in the expression "7:h=p" h does not refer to the height.
14. Monica: "It does not refer to the height. Is it right, Lisa?" 15. Nicolò (*raising his hand*): "Monica, because h refers only to one (*height*), while k…" 16. Lisa: "Both (*the letters*) … (*Nicolò is speaking*)…no, wait! (*to Nicolò*)" 17. Monica: "One at a time" 18. Lisa: "Both the letters are always the height, but h is only for one (*height*) … only for this one (*Lisa goes near the interactive whiteboard to indicate the incision 7 cm height*), while k is valid for all (*the incisions*)." 19. Monica: "k is valid for every incision. (*Stefano is raising his hand*) Stefano?" 20. Stefano: "The first expression … No, I mean: the second expression is more correct than the first. Battista says … where is it? (*Stefano is trying to find Battista's statement*) 'It is evident that dividing by *7*'. It is 'Dividing by *7*', not 'dividing the height'… that is … 21. Monica: "Dividing 7 by …the height"
Dialogue between Monica and Amalia, who observes that Lisa's interpretation of the two expressions is right and declares that, after having listened what Lisa and Nicolò said, she realised that the expression could be interpreted in different ways. Nicolò asks to intervene.
36. Nicolò: "Monica, in the first statement (*he is referring to the first expression*) 7 is divided by the height. Instead, in the second (*expression*) the height is divided by 7!" 37. Monica: "Very good! So … Many times, I realised that many times it is not the same thing. It is necessary to pay attention. It is necessary to think very carefully to what is written. Exchanging, inverting the numbers is not the same thing."
Monica comments on the usefulness of this kind of activity, stressing on the

reflections developed by the students during this discussion.
39. Amalia: "Last time, you (*Monica*) asked us if this thing, this kind of work, was useful. In my opinion, many (*students*) have changed their mind because they said 'Look, now, thanks to these explanations and to all of these … thanks to all of these explanations, I understood what I have to do."
40. Monica: "Lisa, was this activity really useful for you?"
41. Lisa: "Yes"
42. Monica: "Why?"
43. Lisa: "Ahem, because I have never done this kind of work that … together … even if I made a mistake, because I chose A, the fact that the others chose B and explained their motivations 'oppenned' my mind, it opened my mind."
44. Monica: " 'It oppenned your mind', it is ok!"
45. Nicolò: "Monica, it made us understand that first you have to reason, then you can choose."

The process *'establishing where the learners are in their learning'* is central in this lesson: the discussion is planned in order to support the students in making the motivations of their choices explicit. This enables to highlight erroneous ways of reasoning and incomplete explanations, but also to highlight the evolution of students' reasoning, together with the way in which it is influenced by the other students' interventions.

Our analysis of this excerpt will focus on Nicolò and Lisa, two low-achievers.

We may say that Lisa and Nicolò are *activated as owners of their own learning* during the discussion: they ask to correct their initial answers, effectively motivating their new choice (from line 10). The teacher's conduction of the discussion fosters the activation of this strategy by the two students: she is another fundamental agent within this process. The following diagram (Fig. 10) represents the two cuboids that could be, therefore, highlighted within the FaSMEd three-dimensional model at the teacher's level and at the students' level.

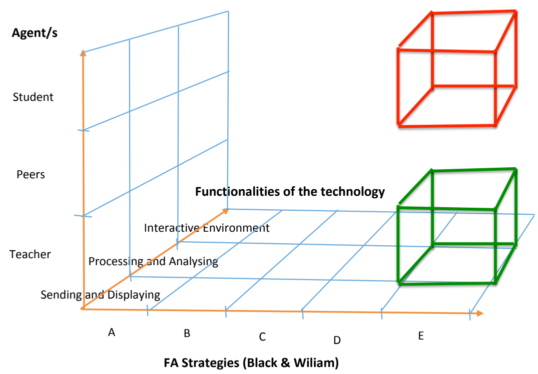

Fig. 10 The activation of the FA strategy "Activating students as owners of their own learning", at the students' and teacher's levels, through the "Processing and Analysing" functionality

There are also evidences of the *activation of students as instructional resources for one another*. Lisa (lines 10 and 18), for example, refers to Alice's intervention (line 2) and elaborates it to start developing her own argumentation. Also Nicolò (line 36) refers to Stefano's intervention (line 20) and elaborates it. Again, we must stress that the teacher is another fundamental agent in this process, because her conduction of the discussion fosters the activation of this strategy at the peers' level. The following diagram (Fig. 11) represents the two cuboids that could be highlighted within the FaSMEd three-dimensional model at the teacher's level and at the peers' level.

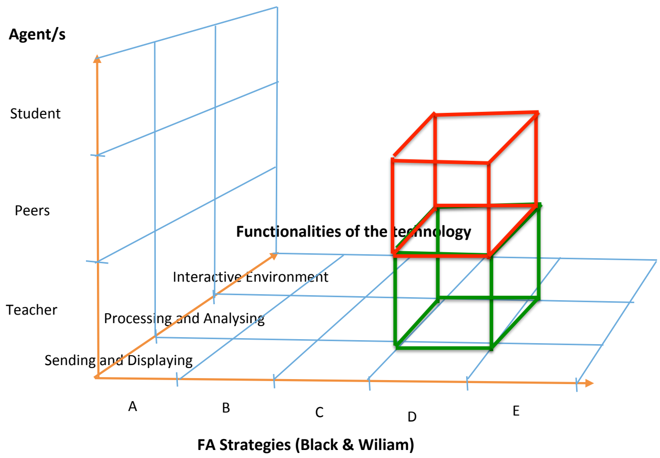

Fig. 11 The activation of the FA strategy "Activating students as instructional resources for one another", at the peers' and teacher's levels, through the "Processing and Analysing" functionality

Another process that is central in this lesson is *'establishing what needs to be done to get them there'*: the teacher intervenes to highlight the most effective ways of reading symbolic expressions and of identifying the one that better represents the involved relations, providing also guidance on how to read the tasks and the texts of the problems (line 37). During the discussion, some interventions are also focused on the positive effects of students' deep involvement in the activities.

It is possible to highlight feedback related to three of the four levels proposed by Hattie and Temperley (2007).

Students' explanations of the reasoning on which their choice was based represent an example of '*feedback about the task'*, which is given *among peers*, because of the different levels of effectiveness of these explanations. For example, Stefano's intervention (line 20), which highlights that the expression 7:h=p does not represent Battista's sentence because, in the expression, 7 is divided by the height and not vice-versa, represents a feedback for Nicolò, who refers to Stefano's statement, clarifying it in an effective way (line 36).

Monica's meta-level intervention in line 37 aims at sharing criteria to correctly identify the expressions that better represent specific relations among quantities: it can be interpreted as '*feedback about the processing of the task*'. This is also an example of the teacher's exploitation of feedback from the students, because Nicolò's statement (line 36) provides Monica the opportunity to discuss the importance of a careful interpretation of symbolic expressions (line 37). Another example of this kind of feedback is Alice's intervention (line 2), which introduces the special case of the 7cm figure, enabling Lisa to understand her mistake and ask to change her answer, proposing motivations (line 10, line 18) that clearly refer to Alice's observation.

The interventions that refer to the importance of listening to each other and of actively participating to class discussions (line 39, line 43, line 45) could be interpreted as '*feedback about self-regulation*'. Amalia's statement (line 39), in fact, provides Monica with the opportunity to discuss on the support given by the discussion with the classmates, asking to Lisa if the activity was useful for her (line 40). It also represents a feedback for Lisa, who proposes a meaningful reflection on the positive effects of the discussion in supporting her understanding of the problem (line 43).

We can therefore highlight, again, two cuboids, within the FaSMEd three-dimensional model (Fig. 12), related to the "Providing feedback that moves learners forward" FA strategy at the teacher's level and at the peers' level).

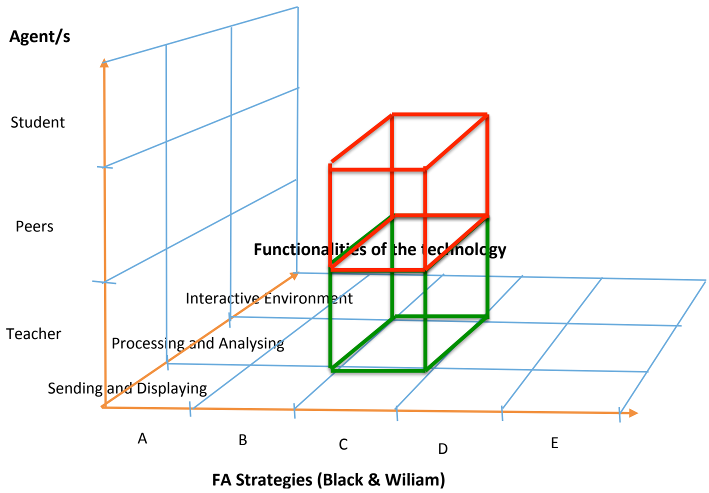

Fig. 12 The activation of the FA strategy "Providing feedback that moves learners forward", at the peers' and teacher's levels, through the "Processing and Analysing" functionality

As already stressed, starting from the poll, Monica has planned a rich discussion, that enabled the activation of different FA strategies by the different participants. This observation enables to highlight another cuboid within the FaSMEd three-dimensional model (Fig. 13), referring to the FA strategy

"Engineering effective classroom discussions and other learning tasks that elicit evidence of student understanding".

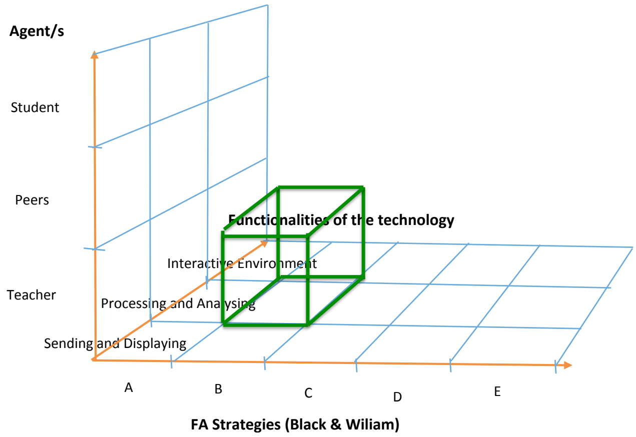

Fig. 13 The activation, by the teacher, of the FA strategy "Engineering effective classroom discussions and other learning tasks that elicit evidence of student understanding", through the "Processing and Analysing" functionality

The technology plays an important role in supporting the agents involved in these processes, in particular in providing feedback to each other. First of all, the software elaboration of the data and the graphical representation of the results of the poll give the teacher the chance to ask for the interpretation of these results and to plan the order of students' interventions during the discussion (Monica decides to start the discussion involving firstly those who have given the wrong answer).

The teacher's choice of not providing students with an immediate automatic correction of their answers may represent a support for students at different levels: (a) it enables to focus on the explanations of the answers, more than on the identification of the correct answer; (b) it pushes the students to motivate their answers; (c) at affective level, the lack of a written evaluation ensures that the students do not feel worried when they comment upon their choices.

Finally the time given to students to choose their answer (in this case, all students answered before the allowed time of 6 minutes) enables them to reflect, in pairs, on the motivations on which their choice is based. The moment that precedes the answer to the poll is, therefore, preparatory to the subsequent discussion.

The effective role played by the chosen functionality of the technology could be also highlighted if we refer to the FaSMEd three-dimensional model to develop a global analysis of the example. The representation of all the cuboids highlighted looking at micro-episodes within our example (Fig. 14), indeed, points out the active involvement of the three agents and the activation of a wide range of FA strategies, drawing attention to the complex dynamics activated through the support of the 'Processing and Analysing' functionality of the technology.

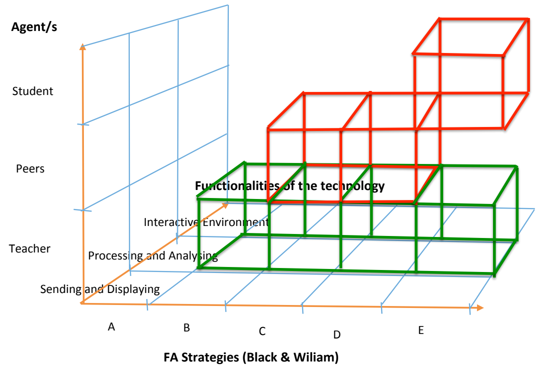

Fig. 14 The use of the FaSMEd three-dimensional model for a global analysis of the example

FA with technology and Professional development

The framework of the MDT, where the interactions between teachers and researchers allow interpretation, at a meta-didactical level, what happened at a didactical level, completes the analysis and illustrates the contribution of the collaborative work both in the design of lessons and for professional development. In the next sections, we present the point of view of teachers through the lens of the MDT model.

FA and PD within the FaSMEd project in France

One of the main concepts of the MDT is the internalisation phenomenon, which has been transversal in the collaborative work between teachers and researchers throughout the project. Particularly, the concept of FA, as collecting data, analysing them, providing feedback to students and therefore adapting teaching according to students' achievement, could be seen as an external component for teachers at the beginning of the project. It has become internal in the observed teachers' practices through the use of technology. Also the use of technology can be considered as an external component becoming internal as a consequence of the role played by the institutional dimension of the project. The process of internalisation leans on the double dialectic where the meta-level consists in conceptualising the FA process, while the didactic level consists in implementing this concept in the classroom practices. In order to illustrate the double dialectic, we will refer to the interviews done with Thomas as well as excerpts of the logbook he filled in all year long.

After Thomas' lessons on linear functions, we interviewed him starting from an analysis of what happened in the classroom and going to more general questions about the way he considers his teaching.

> Researcher: We were in the classroom yesterday when you asked the four fundamental questions relatively to the linear functions […] then you interpreted the results, you showed which questions were well-done or not so well-done and you proposed again some questions...
> Thomas: Where I simplified didactic variables.
> Researcher: Yes and it's very interesting because…
> Thomas: To come back to the mathematical notion.
> Researcher: Yes, to the notion, without the trouble of calculating the sum of fractions and other things...
> […]
> Thomas: On the third quiz we saw the students' results decreasing that was due to two things: the difficulty of calculation with negative numbers and fractions, but also because I made a bad parametrization of the second quiz and the students could do several trials, which makes statistics increase […] this progress was not reliable. Instead, the results of today will give a good representation of the evolution of students' knowledge.

In this excerpt, Thomas remains on the didactic level and analyses the difficulties his students encountered when they took the quiz. He justifies his didactic choices and explains why he eliminates troubles coming from calculation difficulties. In front of each type of task, he takes into account the techniques that students have to know and gives the justification of these choices, completing his praxeology at a didactical level. However, during the interview, interacting with researchers, Thomas moves on to a meta-didactical level when he speaks of the evolution of his FA practices with technology:

> Researcher: You see an evolution but also you see that some students do not succeed yet. Do you plan to do something and how do you plan to continue relatively to this, to the results that you see in the classroom?
> Thomas: To do something... From the beginning of the experimentation, it's much easier with NetSupport. Globally, to give a simple answer... When I use NetSupport, I can intervene individually... I intervene directly with some students and I explain again […] I deal with difficulties, perhaps not of all, but I treat answers individually and now when I use Maple TA, I don't personalize… depending on the statistical results that I get, I decide to give a feedback to the whole class or not.

Thomas has built a technique within a meta-didactical praxeology including technology as a tool allowing to personalize or to redirect his teaching. His reflections at a meta-didactical level are transposed into the *logos* of his didactical praxeologies, becoming the FA principles that justify his FA strategies. This is an example of a dialectic developed at a meta-didactical level that feeds into the dialectic at a didactical level within the MDT process. Thomas justifies his didactical praxeologies, by including the fundamental principles of FA.

Thomas also points out the role of the institutions in his FA practices showing that the global institution of the school does not give time to support low achievers but in the local institution of the classroom, he organizes some special work sessions with students. The particularity is that Thomas leaves to students the responsibility for participating in these supplementary lessons, which contributes to the engagement of students in their own learning. Students' results in the process of FA may influence their decision to participate.

Moreover, as we can read in the teacher's logbook, when he speaks about his praxeologies, he notices some changes that have become stable regarding FA.

> After several experiments, my uses are stable and centred on formative assessment as well as the modification of the students' status of writing.
> Formative assessment: use of NetSupport School and MapleTA.
> Valorisation and exploitation of students' writing (storing data with NetSupport School and processing with the whole class thanks to the IWB)
> The use of software specifically related to mathematics (DGS, Spreadsheets, …) is now regular.

We can interpret Thomas' words as emblematic of an advanced state of the internalisation process concerning the component of FA. Technology emerges as a strong institutional component of his practices that entails teaching modifications relatively to FA. We can detect signs of a shared praxeology in the vocabulary used by Thomas to describe his teaching practices. This shared praxeology as well as the concepts of FA and the functionalities of technology that are stressed by Thomas can be considered as a first result of the whole MDT process. In a general meeting at the end of the year with other teachers involved in the project, Thomas declares: "I had the pleasure of discovering formative assessment. […] I will never go back".

FA and PD within the FaSMEd project in Italy

The teachers collaborating with the Italian team are interviewed after each lesson carried out within the FaSMEd Project. They are asked to reflect on the lesson focusing on:

a. the most effective/most problematic moments during the lesson;
b. the most striking students' interventions, in relation to the given feedback;
c. the effectiveness of the support given by the technology in relation to FA;
d. the support given by the technology to low-achievers.

In this paragraph, we will analyse the reflections carried out by Monica during her interview on the lesson documented in the previous section. We focus, in particular, on Monica's praxeology related to the *task* of developing FA in her class. The aim is to highlight the evolution of her praxeologies during the FaSMEd Project, occurred in the interaction with the researchers' team.

During the interview, Monica focuses on two main aspects: (1) the surprising effective participation of Lisa and Nicolò during the discussion; (2) the functionalities of the technology that better support the students and, in particular, low-achievers.

As regards point (1), Monica stresses that she is really surprised by Lisa and Nicolò's interventions during the discussion:

> Monica: The ones that surprised me more are Lisa and Nicolò. In particular Nicolò, because Lisa is a more logical girl, and if she is focused when doing mathematics, she produces more [than him]. On the contrary, Nicolò faces great difficulties in understanding, even to understand the text of a task. He is the typical student with great difficulties in understanding. The fact that today he managed to tell those things has been a surprise.

In Monica's opinion, Nicolò and Lisa were able to autonomously explain the decision of changing their answers thanks to the discussion activated starting from the screen sharing, where students' answers were displayed. She identified it as one of the most effective moments during the lesson.

As regards point (2), Monica stresses that:

> Monica: …the fact of using this technology and getting the image on the screen allows you, in the meantime, to keep it always present—and when you work in paper and pencil environment, you don't have it, or you get it only on the spot. Then, when the students write and try to answer, many times they make mistakes, they scribble, and so their paper is not clean anymore. Having the possibility, with the tablets, also to delete and get all the screen clean again, so to be able to start again, in my opinion, creates also a mental order not possible in paper and pencil, where, when you write, you cannot delete anymore, you get tired and you lose your focus a lot.
>
> …the possibility of showing the solutions on the screen, the rapidity in being able to see things: they send you their solutions and you can show them to the others. And seeing is not the same as only listening to. In a lesson with paper and pencil worksheets you can listen to all answers and report them on the blackboard, but it takes more time. It takes too much time and the children get lost.

On one side, Monica focused on a functionality of the technology that can be included in the 'sending and displaying' category: in fact, she stresses again that the 'displaying of the screens' is efficient because it enables the students to see, on the interactive whiteboard, the other students' productions, reflecting on them without having to imagine and/or memorise them. On the other side, she focused on another component of the technology: the tablets. She specifically focused on the possibility, given by tablets, of deleting mistakes. In Monica's opinion it supports, in particular, those students (like Lisa and Nicolò, for example) who tend to be not concentrated.

Monica is part of a group of teachers that are involved also in another long-term regional project, called AVIMES[5], aimed at fostering FA in school. Monica and her colleagues are used to collecting students' written productions during

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[5] AVIMES (http://www.avimes.it/who%20are%20we.htm) is a Regional Project focused on research, innovation and professional development in the field of school self-evaluation.

class discussions to share and discuss them with the class. A vision of FA in line with our perspective was, therefore, an internal component of Monica's praxeology also before her participation to FaSMEd. This information enables us to characterize Monica's initial praxeology when she started her experience within the FaSMEd Project: (a) the didactical techniques used by Monica to develop FA in her classes include the sharing of students' productions and collective discussions to make the students compare their works and reflect on them; (b) the theories/justifications of the techniques developed in AVIMES. Digital technologies are external components to this praxeology.

In her interview, Monica compared her way of working during the AVIMES lessons with the new way of working with the support of the digital technology. She stressed the rapidity in which the technology enables collecting, displaying and sharing the students' answers and the consequent saving of time:

> Monica: The fact of having always in real time, on the screen, the various solutions…We, with AVIMES, were used to working in this way: collecting the students' solutions, transcribing what they say and what you say… but it takes a longer time, because you collect them [through the audio-recording], write them down, and then report them. This is a loss of time. With the technology, you can get immediately all projected on the screen and the students' attention and concentration keeps longer.

When asked if she would change something in the lesson, Monica introduced also a reflection on her way of working with this technology. Monica, as many other colleagues, who declare to have faced some difficulties because they were not used to digital technologies during their lessons, seems recognising the effectiveness of the use of a connected classroom technology in fostering FA in her class:

> Monica: No, currently I would not change anything. I'm a little getting used to this. Maybe I could get better results as I were more able to use the computer; if I were quicker I could do something more. Being more able in getting, opening, and closing files, maybe also the students' attention would be greater. A great part depends also on the teacher carrying out the lesson: results do not depend only on technology, but also on me, and maybe before the end of the project I can do something more.

We can therefore highlight an evolution of Monica's praxeology: digital technology, which was external, is slowly becoming an internal component. In fact, Monica recognizes the role it plays, integrated with the other techniques that were already part of her praxeology, in supporting FA in her classes. This evolution is still in progress. The evolution of the *logos* components of the praxeology, does not go at the same pace as the evolution of the technique. In fact, we can notice that Monica does not explicitly refer to the theoretical frame that support the approach developed within FaSMEd. The appropriation of this frame will require further discussion with the teachers and activities specifically devoted to a real sharing of the theoretical basis of this work.

Conclusion

This paper had a twofold aim: (1) highlighting how the different functionalities of technology could enable the enactment of FA strategies at the teacher's, the students' and the peers' levels; and (2) characterising the dynamics that intervene within programs involving a strict collaboration between teachers and researchers.

Concerning the first aim, the analysis of the selected episodes shows clearly the contribution of technology as a medium facilitating the different FA strategies but also the dynamics between these strategies. In particular, the Italian case study points out how the FA strategies and the different levels of feedback emerge in the classroom interactions. The French example highlighted different moments in a sequence, providing a dynamic view of the FA process. The FaSMEd three-dimensional framework enabled us to describe and to analyse the FA lessons from both static and dynamic perspective, considering both the teacher and the student viewpoints. And particularly this model points out the role of technology as a facilitator of the whole process.

Our analysis has highlighted the interrelations between the different functionalities of technology and the FA strategies. The possibility given by technology to store data and the ease to come back to these data is an important functionality that teachers can use to enhance their teaching strategies. As recognised also by the teachers in the interviews, technology is not at the base of FA, but appears as an essential tool to improve the effects of FA for students and for teachers as well. For example, it is possible to *clarify learning intentions and criteria for success* also without technology but technology supports the teacher in making these intentions explicit and to share them with the students, at both individual and class level. Concerning students, the strategies of *activating students as the owners of their own learning* and *activating students as instructional resources for one another* appear as the core of FA, since they enable the active involvement of all the agents (teacher, students, peer/group) within the FA process.

Moreover, the analysis enabled to highlight interrelations between different FA strategies: for instance, *engineering effective classroom discussions and other learning tasks that elicit evidence of student understanding* is facilitated by giving the opportunity of *providing feedback that moves learners forward* on the spot as well as after reflection.

As regards the second aim, the teachers-researchers teams engaged in a process of design-based research benefit of a professional development based on the collaborative work. The Meta-Didactical Transposition framework can show the evolution of beliefs about FA with technology and contribute to the understanding of the meta-level of reflection necessary for a daily use of FA strategies in the classroom. The double dialectic and the internalization of components had a direct impact on the involved teachers but more generally their analysis give information that can be useful for in-training sessions.

The importance of teachers as guides in FA lessons with technology has essential consequences on their practices in terms of professional development. When we began the project, most of the involved teachers stated that FA was present in their practices. However, most of the time, FA was not developed over time and appeared occasionally in the classroom more as a reassuring method than as a teaching strategy. Considering technology as a tool enabling to enhance teaching strategies including FA is surely an important issue of the next years, regarding teachers' professional development. .

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